Solutions to Chapter 9 Exercises

Chaos: An Introduction to Dynamical Systems (Alligood, Sauer, Yorke)

Background used. For $\dot v = f(v)$ with flow F_t, the variational equation is $\dot J_t = A(t)J_t$, $J_0 = I$, where $A(t) = Df(F_t(v))$ and $J_t = DF_t(v)$. Lyapunov exponents of the flow are those of the time-1 map: if $r_1(n) \ge \cdots \ge r_k(n)$ are the semi-axes of the ellipsoid J_n(unit ball), then $h_i = \lim \frac{1}{n} \ln r_i(n)$. Liouville's formula (9.23) gives $\det J_t = \exp \int_0^t \operatorname{Tr} A(s)\, ds$. Since $r_1(n) \cdots r_k(n) = |\det J_n|$, the sum of the exponents is

$$h_1 + \cdots + h_k = \lim_{n\to\infty} \frac{1}{n} \int_0^n \operatorname{Tr} A(s)\, ds.$$

Exercise 9.1

The Lorenz equations are $\dot x = \sigma(y - x)$, $\dot y = rx - y - xz$, $\dot z = xy - bz$. On the z-axis ($x = y = 0$) we have $\dot x = 0$ and $\dot y = 0$, so the z-axis is invariant, and there $\dot z = -bz$. Hence the solution with initial condition $(0, 0, z_0)$ is

$$(x, y, z)(t) = (0, 0, z_0 e^{-bt}),$$

which moves along the z-axis and converges exponentially to the origin as $t \to \infty$ (for $b > 0$). (The z-axis is the one-dimensional part of the stable manifold of the origin; in backward time the solution runs off to infinity along the axis.)

Exercise 9.2

The Chua system is $\dot x = c_1\big(y - x - g(x)\big)$, $\dot y = c_2(x - y + z)$, $\dot z = -c_3 y$, with

$$g(x) = \begin{cases} m_1 x + m_1 - m_0, & x \le -1, \\ m_0 x, & |x| \le 1, \\ m_1 x + m_0 - m_1, & x \ge 1. \end{cases}$$

At an equilibrium (with $c_i \neq 0$), $\dot z = 0$ gives $y = 0$, then $\dot y = 0$ gives $z = -x$, and $\dot x = 0$ gives $g(x) = -x$. Solve in each piece.

- $|x| \le 1$: $m_0 x = -x$, i.e. $(m_0 + 1)x = 0$.
- $x \ge 1$: $(m_1 + 1)x = m_1 - m_0$, i.e. $x = \dfrac{m_1 - m_0}{m_1 + 1}$.
- $x \le -1$: $x = -\dfrac{m_1 - m_0}{m_1 + 1}$.

If $m_0 < -1 < m_1$: in the middle piece $m_0 + 1 \neq 0$ forces $x = 0$. In the outer pieces $m_1 + 1 > 0$ and $m_1 - m_0 > m_1 + 1$ (because $-m_0 > 1$), so $x = \pm\frac{m_1 - m_0}{m_1 + 1}$ has $|x| > 1$, which is consistent with the piece. So there are exactly **three** equilibria: $(0, 0, 0)$ and $\pm\big(\frac{m_1 - m_0}{m_1 + 1},\, 0,\, -\frac{m_1 - m_0}{m_1 + 1}\big)$.

If $m_0 = -1$: the middle equation $0 \cdot x = 0$ holds for every $x \in [-1, 1]$, so every point $(x, 0, -x)$ with $|x| \leq 1$ is an equilibrium: **infinitely many**. (The outer pieces give $x = \pm 1$, the endpoints of this segment.)

Exercise 9.3

For $\ddot{x} + c\dot{x} - x + x^3 = \rho \sin t$ write $\dot{x} = y$, $\dot{y} = -cy + x - x^3 + \rho \sin t$, $\dot{t} = 1$. Then

$$A(t) = \begin{pmatrix} 0 & 1 & 0 \\ 1 - 3x^2 & -c & \rho \cos t \\ 0 & 0 & 0 \end{pmatrix}, \qquad \operatorname{Tr} A = -c.$$

By (9.23) the area-contraction rate per iteration of the time-2π map is $\exp\big(\int_0^{2\pi} (-c)\, dt\big) = e^{-2\pi c}$. The map is dissipative exactly when $e^{-2\pi c} < 1$, i.e. for $c > 0$.

Exercise 9.4

For $\dot{v} = Av$ the Jacobian $A(t) \equiv A$ is constant, so $J_t = e^{At}$ and $\det J_1 = \exp(\operatorname{Tr} A)$. The volume-contraction factor per iteration of the time-1 map is

$$\det e^A = e^{\operatorname{Tr} A},$$

so the system is dissipative iff $\operatorname{Tr} A < 0$.

Exercise 9.5

Write $\dot{x} = y$, $\dot{y} = -cy - P_x(x)$. Then $A(t) = \begin{pmatrix} 0 & 1 \\ -P_{xx}(x(t)) & -c \end{pmatrix}$ has $\operatorname{Tr} A(t) = -c$, so $\det J_n = e^{-cn}$ by (9.23). The image of the unit disk under J_n is an ellipse with semi-axes $r_1(n), r_2(n)$ and area $\pi r_1(n) r_2(n) = \pi |\det J_n|$, so $r_1(n) r_2(n) = e^{-cn}$. Therefore

$$h_1 + h_2 = \lim_{n \to \infty} \frac{1}{n} \big(\ln r_1(n) + \ln r_2(n) \big) = \lim_{n \to \infty} \frac{1}{n} (-cn) = -c.$$

Exercise 9.6

Now $\ddot{x} + P_x(x) = 0$ is the case $c = 0$ of 9.5, so $h_1 + h_2 = 0$. A periodic orbit of the planar system is bounded and contains no equilibrium, so by the consequence of (9.18) (the tangent direction $f(v)$ is neither expanded nor contracted) one exponent is 0. Hence the other is 0 as well: **both Lyapunov exponents are** 0.

Exercise 9.7

If $v(t)$ solves $\dot v = f(v)$, then $w(t) = v(ct)$ solves $\dot w = cf(w)$. So the flow of the second system is $\tilde F_t = F_{ct}$: its time-1 map is the time-c map of the first. After n time units the stretching for the second system is the stretching for the first after cn time units, so (for $c > 0$)

$$\tilde h_i = \lim_{n\to\infty} \frac{1}{n} \ln r_i^{(1)}(cn) = c\, h_i.$$

Thus **each Lyapunov exponent is multiplied by** c; signs (and hence chaos or not) are unchanged. For $c = 0$ all exponents are 0; for $c < 0$ time is reversed as well as rescaled, which turns the exponents into $|c|$ times the negatives of the original ones (when these exist).

Exercise 9.8

Let $u(t)$ be a periodic solution with period T and let $v(t)$ be an orbit with $|v(t) - u(t)| \to 0$ (this is the sense in which an orbit "converges to" a periodic orbit, the flow analogue of Definition 3.3; an equilibrium is the case of period any T). Their variational equations are

$$\dot J = A_v(t)J, \quad A_v(t) = Df(v(t)), \qquad \dot J = A_u(t)J, \quad A_u(t) = Df(u(t)),$$

with A_u T-periodic. Since Df is continuous and $|v(t) - u(t)| \to 0$, we get $A_v(t) - A_u(t) \to 0$.

By Floquet's theorem the fundamental matrix of the second equation is $J_t = P(t)e^{Bt}$ with P T-periodic, bounded and invertible with bounded inverse; the Lyapunov exponents of the periodic orbit are the real parts of the eigenvalues of B (that is, $\frac{1}{T}\ln|\mu_i|$ for the Floquet multipliers μ_i). In the coordinates $z = P(t)^{-1}w$ the first equation becomes

$$\dot z = \big(B + E(t)\big)z, \qquad E(t) = P^{-1}(A_v - A_u)P \to 0.$$

By the classical theorem on asymptotically constant linear systems (Perron–Coppel), the characteristic exponents of $\dot z = (B + E(t))z$ with $E(t) \to 0$ are exactly the real parts of the eigenvalues of B. Since P and P^{-1} are bounded, multiplying by $P(t)$ does not change exponents. Hence the Lyapunov exponents of the orbit of v equal those of the periodic orbit, whenever they exist.

Exercise 9.9

(a) $\dot x = y$, $\dot y = -\sin x$, so $Df = \begin{pmatrix} 0 & 1 \\ -\cos x & 0 \end{pmatrix}$. Along $(0,0)$ the variational equation is $\dot J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} J$, $J_0 = I$.

(b) $J_t = e^{At} = \begin{pmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{pmatrix}$, so $J_1 = \begin{pmatrix} \cos 1 & \sin 1 \\ -\sin 1 & \cos 1 \end{pmatrix}$, a rotation. J_1^n is a rotation by n radians, which maps the unit disk to itself, so both Lyapunov numbers are 1 and the exponents are $\mathbf{0}$, $\mathbf{0}$.

(c) At $(\pi, 0)$, $-\cos\pi = 1$, so $\dot J = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} J$.

(d) $J_t = e^{At} = \begin{pmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{pmatrix}$ and $J_1 = \begin{pmatrix} \cosh 1 & \sinh 1 \\ \sinh 1 & \cosh 1 \end{pmatrix}$. It is symmetric with eigenvalues $e^{\pm 1}$ (eigenvectors $(1,1)$ and $(1,-1)$), so J_1^n has semi-axes e^n and e^{-n}. The Lyapunov numbers are e and e^{-1} and the exponents are $+\mathbf{1}$, $-\mathbf{1}$.

(e) The orbits of the pendulum fall into three types.

- The equilibrium $(0,0)$: exponents $0, 0$ by (b).
- Periodic orbits (oscillations inside the separatrix, and rotations, which are periodic on the cylinder because the equation is 2π-periodic in x): exponents $0, 0$ by Exercise 9.6.
- The saddle $(\pi, 0)$: exponents $+1, -1$ by (d), and the two separatrix (heteroclinic) orbits, which converge to the saddle equilibria $(\pm\pi, 0)$ as $t \to \pm\infty$ and so share its exponents $+1, -1$ by Exercise 9.8 (the tangent direction decays like e^{-t}, consistent with the exponent -1).

So every bounded orbit has exponents $(0,0)$ or $(+1,-1)$. In particular there are no chaotic orbits: the separatrix orbits have a positive exponent, but their ω-limit set is an equilibrium, which Definition 9.6 excludes.